

Compute the segment advantage

$$A_{GRPO} = \frac{1 - \frac{5}{9}}{\frac{\sqrt{14}}{9}} = \frac{4}{\sqrt{14}} \left(\mu = \frac{5}{9}, \sigma = \frac{\sqrt{14}}{9} \right),$$

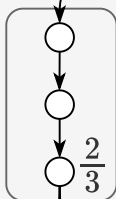
$$A_{RLOO} = 1 - \frac{5}{9} = \frac{4}{9}$$

Prompt
(Question)

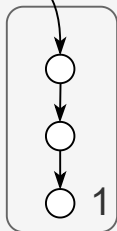
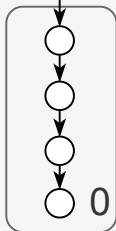
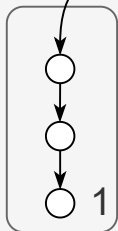
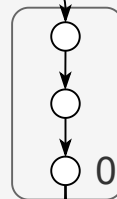
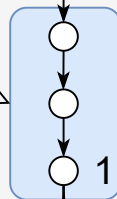
$K = 3$

$$V = (1 + \frac{2}{3} + 0)/3 = \frac{5}{9}$$

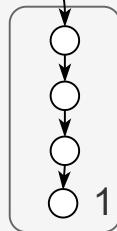
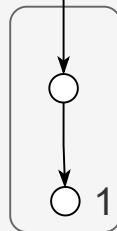
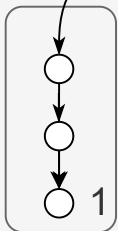
Segment
Group 0



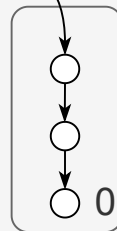
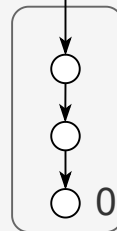
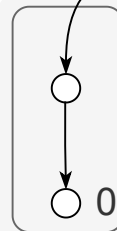
One segment
has M tokens



Segment Group 1



Segment Group 2



Segment Group 3